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AI-Based Food Demand Prediction and Waste Reduction System

IBM SkillsBuild | ML & Applied AI Internship

Submitted by:
Himanshi Nagar

College:
ABES Engineering College, Ghaziabad

Department:
Electrical & Computer Engineering (ELCE)

Project Domain:
Machine Learning and Data Analytics

Sustainable Development Goal:
SDG 12 – Responsible Consumption and Production

Better Planning Today A Greener Tomorrow



LESS FOOD WASTE | SUSTAINABLE FOOD SYSTEMS | HEALTHIER COMMUNITIES

2 Introduction & Problem Statement

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Introduction

Food demand prediction is the process of estimating how much food people are likely to consume over a particular period. Accurate demand estimation can help food service providers make better preparation decisions.

Problem Statement

- Restaumants, cafeterias, hostels and canteens often face difficulties in estimating daily meal demand.
- Preparing excess meals can result in food surplus, while preparing insufficient meals can cause shortages.
- Traditional planning methods may not adequately consider factors such as weather conditions, weekdays, and special events.



Problem to Solve:

Develop a machine learning-based system that estimates meal consumption and supports better food preparation planning.

3 Project Objectives

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1 Predict Meal Demand

Estimate the number of meals likely to be consumed using a machine learning model.

2 Analyze Relevant Factors

Study the relationship between meal consumption and factors such as day of the week, temperature, rainfall and special events.

3 Visualize Data

Present historical meal information and demand predictions through an interactive dashboard.

4 Support Better Planning

Help food service providers make informed decisions about meal preparation.

5 Promote Sustainability

Support responsible food consumption and help reduce potential food surplus, aligning with SDG 12.

12 RESPONSIBLE CONSUMPTION AND PRODUCTION



4 Dataset and Input Features

Dataset Description

- File: food_demand_sample.csv
- Total rows used: 180
- Contains historical meal-related information.

Input Features

Feature	Description
`day_of_week`	Day of the week
`temperature_c`	Temperature in °C
`rainfall_mm`	Rainfall in mm
`special_event`	Indicates special event
`meals_prepared`	Meals prepared
`meals_consumed`	Target variable

Data Preparation

The model uses numerical and categorical preprocessing, including missing-value imputation and one-hot encoding for categorical features.



5 Proposed Methodology

1 Dataset Loading

Load the CSV dataset using Pandas and select the features required for prediction.

2 Data Preprocessing

Prepare numerical and categorical variables and handle missing values using imputation techniques.

3 Train-Test Split

Divide the dataset into training and testing sets (80:20 ratio).

4 Random Forest Regression

Train a Random Forest Regressor using Scikit-learn to estimate the number of meals consumed.

5 Model Evaluation

Evaluate using MAE, RMSE and R^2 score.

6 Streamlit Dashboard

Integrate the saved model into a Streamlit app for user input and prediction.

Evaluation Metrics: MAE, RMSE, R^2 score

6 Technology Stack

Python

Main programming language

Pandas

Data loading, processing and analysis

Scikit-learn

Machine learning model training & evaluation

Streamlit

Interactive web dashboard

Joblib

Model saving and loading

CSV

Dataset format

GitHub

Code repository

System Components

- Dataset
- Model training script
- Saved model
- Streamlit application

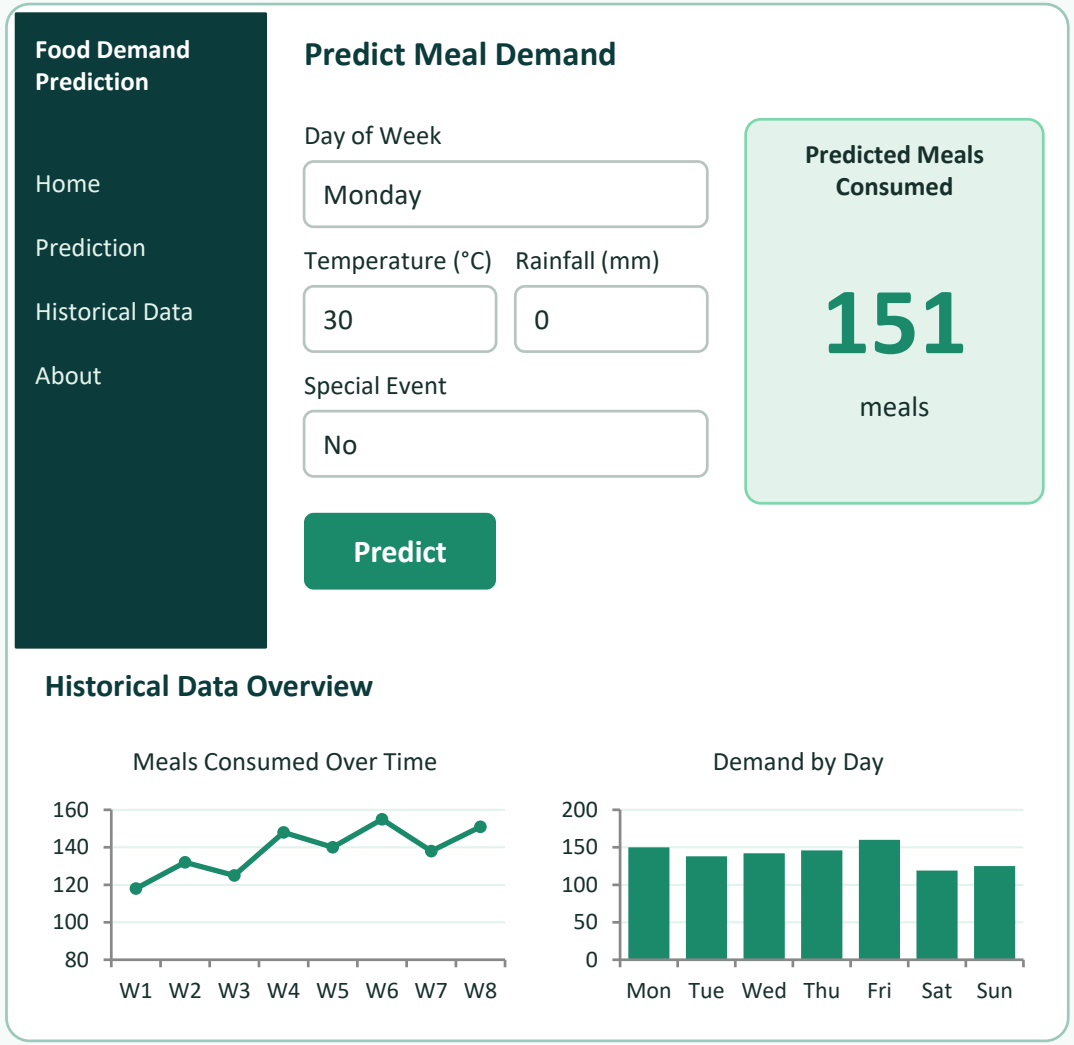
7 Results and Dashboard

Model Evaluation Results

MAE	RMSE	R^2 Score
9.51 meals	11.63 meals	0.554 score

Example Predictions

Monday, 30°C, no rain, no special event	→ ~151 meals
Saturday, 35°C, 10 mm rainfall, special event	→ ~119 meals



8 Conclusion and Future Scope

Conclusion

- Developed a working prototype for predicting meal consumption.
- Integrated the trained model with a Streamlit dashboard.
- Demonstrated how data-driven estimates can support food preparation planning.
- Supports SDG 12 by exploring a data-driven approach to responsible food consumption and reducing potential food surplus.

Future Scope

- Train and validate the model using real-world data.
- Improve prediction performance through model tuning.
- Integrate a database for storing daily meal records.
- Deploy the dashboard online for easier access.
- Track actual surplus and food waste to measure real-world impact.

Thank You!

Questions and Discussion